

Project Plan – Intelligent Trip Planning Assistant

Name: Zhiqian gu **Date:** 13 August 2025 **Pathway:** Back-end First Developer

1. Research Problem

Modern travel planning often requires users to juggle multiple platforms (flights, hotels, attractions, transportation), leading to fragmented itineraries, inefficient routes, and plans that are not personalised to individual needs. Existing trip planners frequently lack:

1. Integrated real-time data (e.g., weather, opening hours, disruptions)
2. Advanced personalisation that adapts to evolving user preferences
3. Optimised routing balancing travel time, cost, and sustainability factors

This Research aims to design and implement a backend-driven intelligent travel planning system and a minimalist yet fully functional traditional web interface (HTML + CSS + JavaScript). By leveraging multiple APIs, secure authentication, caching technology, and a scalable architecture, the system will provide users with personalized, context-aware, and optimized itinerary arrangements.

2. Literature Review Keywords:

- Intelligent trip planning systems
- Intelligent travel recommendation systems
- Context-aware recommendation algorithms
- Route optimisation algorithms (e.g., Dijkstra, A* search)
- Personalisation in recommender systems
- Real-time data integration in trip planning
- Java Spring MVC back-end architecture
- Redis caching for high-performance APIs
- JWT-based authentication and security

3. System Architecture:

Frontend (HTML + CSS + JavaScript)

↓ HTTP

Backend API (Java + Spring MVC)

↓ MyBatis XML Mapper

Database Layer (MySQL + Redis Cache)

↓ External APIs (Geoapify, OpenStreetMap, Weather API)

3. Core Data Structures

Component	Data Structure
User Profile	JSON (MySQL Table)
Itinerary	Relational Table
POI Data	Relational Table
Real-time Cache	Key-Value (Redis)

Storage Design:

- **Relational DB (MySQL)** for user profiles, itineraries, POI data
- **Redis** for caching frequently accessed routes and POI queries
- **JWT token store** for stateless authentication

Intended Interface:

The system will be accessible through a traditional web interface featuring:

- HTML + CSS + JavaScript front-end for itinerary search, display, and basic interactivity
- RESTful APIs (Spring MVC) for itinerary generation, POI search, and live updates
- Responsive layout for desktop and mobile browsers
- API security with JWT authentication
- Caching with Redis to ensure performance
- AI-based itinerary recommendations (model to be decided later), like LLM

Backend Architecture:

- **Data Ingestion Layer** – integrates with OpenStreetMap, and weather APIs
- **Core Engine** – combines user preferences, route optimisation algorithms, and real-time updates
- **Redis Cache Layer** – accelerates repeated queries and reduces API latency
- **Authentication Layer** – JWT-based stateless API authentication
- **API Layer** – handles requests, returns JSON for front-end rendering
- **MyBatis XML Mapper** – structured SQL mapping for maintainable database access

Target Grade Tier:

- **GPA: 6**